U2

miércoles, 24 de septiembre de 2025 10:00 a. m.

U2S1: Sequential Clustering Algorithms

domingo, 21 de septiembre de 2025
08:52 p. m.

What are they?
Straight forward fast methods that produce single clustering

Characteristics:

- In most: all feature vectors will only be presented only once.
- Result depends on the order the vectors were presented
- Produce compact hyper spherical or hyper-ellipsoidally shaped clusters.
- Number of clusters is not known apriori

Standard Notation

- $d(x, C)$ distance or dissimilarity feature vector x and the set (cluster) C

Parameters:
threshold: θ
max # of clusters q

BSAS: Basic Sequential Algorithmic Scheme

1. $m = 1$ (m = # of clusters the algorithm has made so far)
2. $C_m = \{x_1\}$
3. For $i = 1, 2, \dots, N$ ← the whole set
   - Find $C_k : d(x_i, C_k) = \min_{1 \le j \le m} d(x_i, C_j)$ (the cluster that is closest to the vector)
   - If $d(x_i, C_k) > \theta$ and $m < q$ then (if it is further than θ and quota isnt met)
     - $m = m+1$
     - $C_m = \{x_i\}$
   - Else
     - $C_k = C_k \cup \{x_i\}$
     - When necessary update representatives
   - End if

IDEA: Each new vector is assigned either to
- an existing cluster
- a new cluster

depending on its distance from the already formed ones.

NOTES
- Order of the vectors alters the cluster outcomes.
- Choices of the treshold affect the # of clusters made
  - θ small → unesessary clusters
  - θ big → less than enough clusters
- q we chose it. This constraint may be set or we allow the algorithm to choose.

REMARKS: Different distance measures $d(x, C)$ will lead to different algorithms // When C is represented by a point m_C $d(x, C) = d(x, m_C)$ // If we are working with similarity we do max instead of min. // Make compact clusters not suitable for complex.

COMPUTATIONAL COMPLEXITY: $O(n)$ goes only through all the data only once.

MBSAS: Modified Basic Seavencial Algorithm Scheme

- $m = 1$
- $C_m = \{x_1\}$
- For $i = 2, \dots, N$
  - Find $C_k : d(x_i, C_k) = \min_{1 \le j \le m} d(x_i, C_j)$
  - If $d(x_i, C_k) > \theta$
    - $m = m+1$
    - $C_m = \{x_i\}$
  - End if
- End for

Part 2 pattern classification
- For $i = 1, \dots, N$
  - If x_i has not been assigned
    - Find $C_k : d(x_i, C_k) = \min_{1 \le j \le m} d(x_i, C_j)$
      - $C_k = C_k \cup \{x_i\}$
      - Where necessary, make rep updates.

CLUSTER NUMBER ESTIMATION

Method to do q estimation
For $\theta = a$ to b with steps $= c$
run BSAS with each θ to get m
get the m mode or stability.

Then get q from said m.
It can be better to get θ with the elbow
the flat region was stability.

MBSAS:
Here we have a better on the BSAS that assings before knowing the final cluster formation. MBSAS overcomes this by going through all vectors 2 times

Basic IDEA:
First phase: determination of clusters.
Second phase: presented the unassigned to make final clusters

ITERATIVE MEAN VECTOR

When C is represented by a single vector $d(x, C)$ becomes
$$d(x, C) = d(x, m_C)$$
m_C is a representative of C
we know how to update.

$$m_C\ new = \frac{(n_{C_k}^{new} - 1)\, m_{C_k}^{old} + x}{}$$

we know how to update.

$$m_c\,new = \frac{(n^{new}_{C_k} - 1)\, m^{old}_{C_k} + X}{n\, C^{new}_{k}}$$

U2S1: Hierarchical Algorithms

martes, 23 de septiembre de 2025
01:10 p. m.

$\boxed{\text{NESTED CLUSTERING}}$ Note that at least one cluster of R_1 is a proper subset of R_2 Example:

$$R_1 = \{\{x_1, x_3\}, \{x_4\}, \{x_5, x_2\}\}$$

is nested in

$$R_2 = \{\{x_1, x_2, x_3\}, \{x_5, x_2\}\}$$

counter example: R_1 is not nested in R_3 & R_4

$$R_3 = \{\{x_1, x_4\}, \{x_3\}, \{x_2, x_5\}\}$$

$$R_4 = \{\{x_1, x_2, x_4\}, \{x_3, x_5\}\}$$

Aglomerative Algorithms: Initial state is N clusters one V elemen ∈ X

- first step, clustering R1, is produced and must contain N-1 sets such that $R_0 \sqsubset R_1$
- This procedure will go on until R_{N-1} that will form a single cluster the set of all X.
- Notice hierarchy: $R_0 \sqsubset R_1 \sqsubset \ldots \sqsubset R_{N-1}$

Divisive Algorithms: They follow the inverse path just described.

- Initial cluster, R_0, is a sing set all of X.
- First step is R_1 must be 2 sets $R_1 \sqsubset R_0$
- This is done until R_{N-1} which is n sets one per each element of X
- Note that $R_{N-1} \sqsubset R_{N-2} \sqsubset \ldots \sqsubset R_1 \sqsubset R_0$

$\boxed{\text{HIRERCHICAL CLUSTERING}}$ This algorithm has a different philorophy prom BSAS it produces not a partion but a hierarchy of clustering

Partition Clustering | Hierarchical Clustering

Hierarchical clustering Algorithms.

produce → hierarchy of nested clustering

They take N steps: as many as the number of data vectors.

At each of the N steps: step t, a new clustering is obtained based on the clustering obtained at t-1.

These algorithms may be:

AGLOMERATIVE | DIVISIVE

junta | separa.

U2S2: Agglomerative Algorithms

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$\boxed{\text{GAS: Generalized Agglomerative Scheme}}$

let $C(C_i, C_j)$ be a function defined for all the possible pairs of clusters of X

This function gives the proximity between cluster C_i & C_j

let t denote the level of hierarchy.

GAS: Generalized Agglomerative Scheme

cluster C_i & C_j

let t denote the level of hierarchy

- **Initialization:**
  - Choose $R_0 = \{C_i = \{x_i\}, i = 1, \dots, N\}$
  - t_0

This is R_0 where each data point is its own cluster

- **Repeat:**
  - $t = t + 1$
  - Among all pairs on (C_r, C_s) R_{t-1} find (C_i, C_j) that satisfies:
    - if g is the dissimilarity function $g(C_i, C_j) = \min_{r,s} g(C_r, C_s)$

    (use max if we are working on similarity)
  - Define $C_q = C_i \cup C_j$ and produce a new clustering $R_t = (R_{t-1} - \{C_i, C_j\}) \cup \{C_q\}$

    (we get rid of the ones we merged and add the merged result)

Main idea: on each iteration the new R_t will only differ on 1 cluster as it will only merge 2 we found closest.
This allows for the one be nested in all successive clusters
$R_{t1} \subset R_{t2}$, $t1 < t2$, $t2 = 1, \dots, N-1$

AKA once 2 things are in the same cluster they will be for all the next hierarchies.

WEAKNESS OF THE METHOD:

- if a poor clustering choice is made earlier it is not possible to recover
- The algorithm goes over the data N^3 times but the actual big O depends on how we are defining in metric & method the distance between 2 clusters.

PATTERN MATRIX:

This means $D(X)$ is the $N \times l$ matrix whose ith row is the transposed ith vector of X

$$\begin{array}{cc} & l\text{ - dimentions} \\ \begin{matrix} N \\ \text{vectors} \end{matrix} & \begin{bmatrix} X_1 = \{X_{11}, X_{12} \dots X_{1l}\} \\ X_2 = \{X_{21} \dots X_{2l}\} \\ \vdots \quad \dots \end{bmatrix} \end{array}$$

Similarity / Dissimilitary matrix $P(X)$ is an $N \times N$ matrix whose $[i,j]$ elemen is the distance between the ith and jth elements of X we may call it PROXIMITY MATRIX to include both cases.

$$\begin{array}{cc} & N \\ N & \begin{bmatrix} 11 & \dots & N1 \\ & \ddots & \vdots \\ 1N & & NN \end{bmatrix} \end{array}$$

In general the PROXIMITY MATRIX is symmetric

Moreover:
- If P is a sim diagonal $= s_0$
- If P is a diss diagonal $= d_0$

EXAMPLE OF PATTERN & PROXIMITY

let $X = \{x_i, i = 1, \dots, 4\}$ with
$x_1^T = (1,1)$
$x_2^T = (2,1)$
$x_3^T = (5,4)$
$x_4^T = (6,5)$

PATTERN

$$D(X) = \begin{bmatrix} 1,1 \\ 2,1 \\ 5,4 \\ 6,5 \end{bmatrix}$$

THE DENDOGRAM:

- A threshold dendogram is a way to represent
  - Used to show clusters formed.
- Cut a dendogram and you get a partition.

Dissimilarity matrix

L_1: Manhattan

	x_1	x_2	x_3	x_4
x_1	0	1	7	9
x_2	1	0	6	8
x_3	7	6	0	2
x_4	9	8	2	0

L_{max}: Chebyshev

	x_1	x_2	x_3	x_4
x_1	0	1	4	5
x_2	1	0	3	4
x_3	4	3	0	1
x_4	5	4	1	0

Agglomerative Methods Matrix Theory

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04:34 p. m.

These are really particular cases of GAS

Input:
- $N \times N$ prox matrix
- $P_0 = P(X)$ derived from X

At each level t, the size of the matrix becomes: $(N-t) \times (N \times t)$

P_t follows P_{t-1} by

DELETING: 2 rows & colums of the merged clusters.

ADDING: a new row & new col for the new cluster.

Matrix Updating Algorithm Scheme (MUAS)

R_0 initial partition
$P_0(X)$ PATTERN MAT
$t = 0$

- Initialization:
  - $R_0 = \{\{\mathbf{x}_i\}, \ i = 1, \dots, N\}$
  - $P_0 = P(X)$
  - $t = 0$

First loop:
add hierarchy $t+1$
find closest cluster pair

- Repeat:
  - $t = t + 1$
  - Find C_i, C_j such that

$$d(C_i, C_j) = \min_{r,s=1,\dots,N, \ r \neq s} d(C_r, C_s)$$

Merge said pair to 1
Update the partition R_t

  - Merge C_i, C_j into a single cluster C_q and form $R_t = (R_{t-1} - \{C_i, C_j\}) \cup \{C_q\}$
  - Define the proximity matrix P_t from P_{t-1}

Merge said pair to 1
Update the partition R_t

Update Algorithms:

$$d(C_i, C_j) = \min_{r,s=1,\dots,N,\ r\neq s} d(C_r, C_s)$$

- Merge C_i, C_j into a single cluster C_q and form $R_t = (R_{t-1} - \{C_i, C_j\}) \cup \{C_q\}$
- Define the proximity matrix P_t from P_{t-1}

[illegible]: a new row & new col for the new cluster.

- Until R_{N-1} clustering is formed, that is, all vectors lie in the same cluster.

The updating is the method to get the distance between 2 clusters

Each method may alter the way the clusters are formed.

Update Algorithms

- The **single link algorithm**:

$$d(C_q, C_s) = \min\{d(C_i, C_S), d(C_j, C_s)\}$$

The single link has a tendency to favor **elongated clusters**.

- The **complete link algorithm**:

$$d(C_q, C_s) = \max\{d(C_i, C_S), d(C_j, C_s)\}$$

The complete link algorithm proceeds by recovering small **compact clusters**.

- The **Ward** or **minimum variance** algorithm. Here, the distance between two clusters C_i and C_j, d_{ij}, is defined as a weighted version of the squared Euclidean distance of their mean vectors, that is,

$$d_{ij} = \frac{n_i n_j}{n_i + n_j} d_{ij}$$

where $d_{ij} = \|m_i - m_j\|^2$.

• Single link: takes min la distancia entre 2 clusters es el menor de esos clusters a C_s.
• Complete: max d to C_s.

WARD or min variance

$$d_{ij} = \frac{n_i n_j}{n_i + n_j} d_{ij}$$

$d_{ij} = \|m_i - m_j\|^2$

In wards mv method
ns are the cardinality of the clusters (sets)

m_s are the centroids (mean vectors.)

When to use

Method	Data trait/Cluster aim
SL	Irregular shape clusters. Non convex shapes. only needs 1 close pair to bundle together.
CL	Compact and spherical. assumes clusters are tight and equidistant.
WARD	1 Clusters are similar in size & density. Optimizes intra cluster variance
	2 When data is very high dimentional. Other 2 methods become too unstable.

Divisive Algorithms

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09:37 a. m.

Divisive algorithms are inverse from agglomerative.

The t-th clustering contains $t+1$ clusters.

In the sequel C_{tj} will denote the j-th cluster of the t-th clustering R_t, $t = 0, \dots, N-1$, $j = 1, \dots, t+1$.

Let $g(C_i, C_j)$ be a dissimilarity function defined for all possible pairs of clusters.

The initial clustering R_0 contains only the set X, that is, $C_{01} = X$.

To determine the next clustering, we consider **all possible pairs** of clusters that form a partitioning of X.

Among them we choose the pair, denoted by (C_{11}, C_{12}), that **maximizes** g.[1]

[1] It can be used a similarity function. In that case we should choose the pair of clusters that minimizes g.

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These clusters form the next clustering R_1, that is, $R_1 = \{C_{11}, C_{12}\}$.

At the next time step, we consider all possible pairs of clusters produced by C_{11} and we choose the one that maximizes g. The same procedure is repeated for C_{12}.

Assume now that from the two resulting pairs of clusters, the one originating from C_{11} gives the larger value of g. Let this pair be denoted by (C_{11}^1, C_{11}^2).

Then the new clustering, R_2 consists of C_{11}^1, C_{11}^2, and C_{12}.

Relabeling these clusters as C_{21}, C_{22}, C_{23}, respectively, we have

$$R_2 = \{C_{21}, C_{22}, C_{23}\}$$

Carrying on in the same way, we form all subsequent clustering.

- Initialization
  - Choose $R_0 = \{X\}$ as the initial clustering.
  - $t = 0$
- Repeat
  - $t = t + 1$
  - For $i = 1$ to t
    - Among all possible pairs of clusters (C_r, C_s) that form a partition of $C_{t-1,i}$, find the pair $(C^1_{t-1,i}, C^2_{t-1,i})$ that gives the maximum value for g.
  - Next i
  - From the t pairs defined in the previous step choose the one that maximizes g. Suppose that this is $(C^1_{t-1,j}, C^2_{t-1,j})$.
  - The new clustering is

$$R_t = (R_{t-1} - \{C_{t-1,j}\}) \cup (C^1_{t-1,j}, (C^2_{t-1,j})$$

  - Relabel the clusters of R_t.
- Until each vector lies in a single distinct cluster.

Different choices of g give rise to different algorithms.

One can easily observe that this divisive scheme is computationally very demanding, even for moderate values of N. This is its **main drawback**, compared with the agglomerative scheme.

Thus, if these schemes are to be of any use in practice, some further computational simplifications are required:

- One possibility is to make compromises and not search for all possible partitions of a cluster.

Split criterion

Let C_i be an already formed cluster. Our goal is to **split** it further, so that the two resulting clusters C_i^1 and C_i^2, are as **dissimilar** as possible.

Initially, we start with

$$C_i^1 = \emptyset \qquad \wedge \qquad C_i^2 = C_i$$

Then, we identify the vector in C_i^2 whose **average dissimilarity** from the remaining vectors is maximum, and we move it to C_i^1.

In the sequel, for each of the remaining $x \in C_i^2$, we compute its average dissimilarity with the vectors of C_i^1:

$$g(x, C_i^1)$$

as well as its average dissimilarity with the rest of the vectors in C_i^2:

$$g(x, C_i^2 - \{x\})$$

Split criterion

If for every $x \in C_i^2$, it holds

$$g(x, C_i^2 - \{x\}) < g(x, C_i^1)$$

then we stop.

Otherwise, we select the vector $x \in C_i^2$ for which the difference

$$D(x) = g(x, C_i^2 - \{x\})) - g(x, C_i^1)$$

is **maximum** (among the vectors of C_i^2, x exhibits the maximum dissimilarity with $C_i^2 - \{x\}$ and the maximum similarity to C_i^1) and we move it to C_i^1.

The procedure is repeated until the termination criterion is met.

One important issue in hierarchical algorithms is determining the best clustering within a given hierarchy. This is equivalent to identifying the **number of clusters** that **best fits** the data.

Lifetime of a cluster

An intuitive approach is to search in the proximity dendrogram for clusters that have a **lifetime**.

The lifetime of a cluster is defined as the absolute value of the difference between the **proximity level** at which it is created and the proximity level at which it is absorbed into a larger cluster.

However, **human subjectivity** is required to reach conclusions.

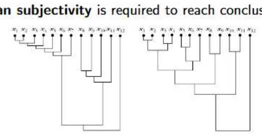

Figure: Example

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Method I

This is an **extrinsic method**, in the sense that it requires determination of the value of a **specific parameter by the user**.

It involves the definition of a function $h(C)$ that measures the **dissimilarity** between the vectors of the same cluster C. That is, we can view it as a **self-similarity measure**.

For example,

$$\begin{aligned} h_1(C) &= \max\{d(x,y),\ x, y \in C\} \\ h_2(C) &= \text{med}\{d(x,y),\ x, y \in C\} \end{aligned}$$

When d is a metric distance, $h(C)$ may be defined as

$$h_3(C) = \frac{1}{2|C|} \sum_{x \in C} \sum_{y \in C} d(x, y)$$

Let θ be an appropriate **threshold** for the adopted $h(C)$.

Then, the algorithm terminates at the R_t clustering if

$$\exists C_j \in R_{t+1} : \ h(C_j) > \theta$$

In words, R_t is the **final clustering** if there exists a cluster C in R_{t+1} with dissimilarity between its vectors greater than θ.

Sometimes the threshold θ is defined as

$$\theta = \mu + \lambda\sigma$$

where μ is the **average distance** between any two vectors in X and σ is its **variance**.

The parameter λ is a **user-defined parameter**.

Thus, the need for specifying an appropriate value for θ is transferred to the choice of λ. However, λ may be estimated more reasonably than θ.

Method II

This is an **intrinsic method**; that is, in this case only the structure of the data set X is taken into account.

According to this method, the final clustering R_t must satisfy the following relation:

$$d_{\min}^{ss}(C_i, C_j) > \max\{h(C_i), h(C_j)\}, \qquad \forall C_i, C_j \in R_t$$

In words, in the final clustering, the **dissimilarity between every pair of clusters** is larger than the **self-similarity** of each of them.

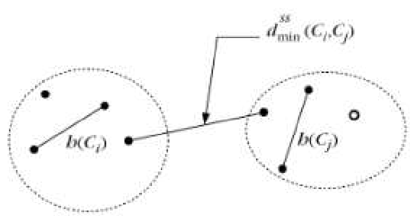

Figure: of the termination condition for method II